COURSE 2 · SECTION 10 · WORKED SOLUTION

DFMEA risk before and after

Section 10 · DFMEA risk view

DESIGN QUESTION

Which design risks were reduced by the recommended actions, which ones are still high, and which high-severity failure modes have no action at all?

Steps

1. Open Section11_Start.pbix (or Engxtech_S10.pbix) > File > Save as > Engxtech_S10_Exercise.
2. DFMEA page: cards Total RPN Before, Total RPN After; column chart LineID by RPN Before and RPN After.
3. Right-click _Measures > New measure, twice (below). Cards for both.
4. The 'high S, no action' table: add DFMEA[O], DFMEA[D] and DFMEA[Action].
5. Matrix: Rows DFMEA[S], Columns DFMEA[O], Values Count of LineID; the S = 9 row total.

DAX from the video

```
DFMEA[RPN Before] = DFMEA[S] * DFMEA[O] * DFMEA[D]
DFMEA[RPN After]  = DFMEA[S_after] * DFMEA[O_after] * DFMEA[D_after]
DFMEA[HighS NoAction] = DFMEA[S] >= 7 && ( ISBLANK ( DFMEA[Action] ) || DFMEA[Action] = "" )
Total RPN Before = SUM ( DFMEA[RPN Before] )
Total RPN After  = SUM ( DFMEA[RPN After] )
```

DAX new in this exercise

```
Lines RPN Before >= 100 = COUNTROWS ( FILTER ( DFMEA, DFMEA[RPN Before] >= 100 ) )
Lines RPN After >= 100  = COUNTROWS ( FILTER ( DFMEA, DFMEA[RPN After] >= 100 ) )
```

DF-01 (seat ring wear) drops from 280 to 72 after the tighter OD tolerance in ECR-3200, because O and D were re-scored; the field ppm in Section 9 is still an early signal, so treat the new O as a forecast. DF-04 (S7) and DF-07 (S8) have no action and keep their RPN.

Two scoring questions to raise in a review: DF-02 'Interference fit check added' is a detection action, yet O drops 4 to 3; a detection action should not lower occurrence. DF-08 'Grommet added, rev B' is a design change, yet D drops 6 to 4; design changes usually do not lower detection. RPN also hides severity (a 9 x 2 x 4 line scores lower than a 6 x 4 x 5 one), which is why the AIAG-VDA FMEA Handbook replaces RPN ranking with Action Priority (AP).

Expected result

Check	Result
DF-01 RPN before / after	**280 / 72**
Sum of RPN before / after	**1470 / 788**
Lines with RPN of 100 or more, before / after	**8 / 2**
High severity (S at least 7) with no action	**DF-04, DF-07**
Lines with S = 9	**3**